AEGIS: Matrix-Free Diagnostics for the Adversarial Fault Lines of Graph Neural Networks

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***Abstract*—Graph neural networks now inform safety-critical decisions in drug-interaction screening, fraud detection, and power-grid contingency analysis, where adversarial perturbations of the graph can flip predictions. Auditing such a model means mapping its *adversarial fault lines*: the most sensitive first-order perturbation direction, per-edge equilibrium-sensitivity rankings, and per-node sensitivity radii. Existing tools supply at most one. AEGIS produces all three from a single object, the *constrained sensitivity matrix* S_c, computed matrix-free through a Neumann-series resolvent and randomized SVD that scale to $N{=}7{,}650$ on one GPU. For the spectral-norm-constrained IGNN subclass we prove a closed-form three-regime characterisation with critical budget $\varepsilon_{\mathrm{crit}}{=}(1{-}\kappa)/\left\|W\right\|_2$ (rate sharp in the all-active case), which trained models satisfy with 2–4× margin. The construction extends to any continuous-edge-weight message passing (GCN, SAGE, GIN, APPNP, IGNN, edge-weighted GAT†): its continuous-to-discrete bridge is positive in *all* $39/39$ cells under the edge-weighted ranking of Proposition 3 (median $\tau{=}{+}0.99$, $p{<}10^{-5}$), beating an edge-weight baseline by $\Delta\tau$ up to $+0.65$ on a fraud graph and reaching $+0.996$ on Amazon Photo ($N{=}7{,}650$), while one query recovers the direction 50-step PGD finds. Masking the top edges by v_{ij} cuts $\sigma_1(S_c)$ damage by $42{\pm}8\%$ (versus 11% for random masking) and survives adaptive recomputation. On IEEE power flow, S_c recovers N-1 rankings on case14–118 (P@10=0.66–0.81) from a learned surrogate, without the admittance matrix exact LODF requires.**

***Index Terms*—Graph neural networks, adversarial robustness, constrained sensitivity, implicit graph neural networks, randomized SVD, power-flow contingency screening**

I. INTRODUCTION

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection via perturbed transaction edges [1], drug-interaction models misfire under perturbed molecular graphs [2], and power grids can miss critical contingencies when their graph representations are structurally fragile [3]. A practitioner deploying a GNN faces a question current tools leave unanswered: *which edges, if perturbed, would cause predictions to fail, and by how much?*

Each existing thread maps only part of these *adversarial fault lines*. Structural attacks [1], [4], [5], [6] rank edges by surrogate gradients but yield no continuous direction and no per-node budget; smoothing [7], [8], [9], [10] and IBP-style certifiers [11], [12] return per-node certificates but no edge ranking or direction; robust-architecture defenses [13], [14], [15] harden models without surfacing per-edge vulnerability; and sober-look critiques [16], [17] flag evaluation pitfalls. No single method returns all three at once. AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) does, from one analytical object: the constrained sensitivity matrix S_c (section IV). Intuitively, S_c records how strongly each edge can push the model's equilibrium prediction, so its leading singular direction, its column norms, and its per-node margins yield the three diagnostics in a single pass. Here "one query" means one matrix-free S_c construction (a single randomized SVD over the equilibrium resolvent), not three separate analyses. The closed-form regime characterisation (Theorem 1) is established for contractive implicit GNNs, and S_c extends to any GNN with continuous edge-weight-modulated message passing (section V-F); the threat model targets edge deletion and weight perturbation (section II).

Contributions. (1) Constrained sensitivity operator. S_c specialises equilibrium IFT sensitivity [18], [19] to *structural* edge perturbations via the projection P_c. One matrix-free query then reads the SVD-optimal direction, per-edge rankings, and per-node radii together, where no prior object yields all three; it matches a dense σ_1 to 0.03% at $N{=}200$ and scales to $N{=}7{,}650$. **(2) Phase-transition theory.** A worst-case three-regime characterisation with closed-form critical budget $\varepsilon_{\mathrm{crit}}{=}(1{-}\kappa)/\left\|W\right\|_2$ for the spectral-norm-constrained IGNN subclass. Trained IGNNs satisfy it with 2–4× margin across our $\kappa_{\max}$ sweep, a provable structural-perturbation safety threshold at $\sim 6\%$ accuracy cost (Theorem 1 and section V-D). **(3) Empirical evaluation.** 10 datasets, 7 architectures, 5 domains (390 runs): four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD [5], AGNNCert [11], and LODF (table III), continuous-to-discrete transfer (fig. 7), and a power-grid N-1 case study (section VI).

II. PRELIMINARIES AND THREAT MODEL

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes and adjacency $A \in \mathbb{R}^{N\times N}$, node features $X \in \mathbb{R}^{N\times d_{\mathrm{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A{+}I)D^{-1/2}$ (self-loops included). Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i-th singular value, and $\mathrm{vec}(\cdot)$ the column-major vectorization. The sensitivity matrices (S, the constrained S_c, the unrolled S_K, and the per-node block S_v) and the scalars that govern them (contractivity $\kappa{=}\left\|J_z\right\|_2$, spectral radius $\rho(J_z)$, pseudospectral index $\eta \geq 1$, critical budget $\varepsilon_{\mathrm{crit}}$, per-node radius r_v) are each defined where they first appear.

IGNN and IFT. AEGIS analyses models that settle to an equilibrium, because that fixed point is what exposes structural sensitivity in closed form. A DEQ-GNN [20], [21] iterates a weight-tied F_θ to a fixed point; the IGNN instantiation [22] uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\mathrm{proj}}), \tag{1}$$

with $W \in \mathbb{R}^{d\times d}$ spectral-normalized [23] so $\kappa \leq \left\|\hat{A}\right\|_2 \|W\|_2 < 1$, ReLU ϕ, and a learned projection X_{proj}. Training uses implicit differentiation through the fixed point [19], [24] with optional Jacobian regularization [25], and a linear head $f(z^*) = W_{\mathrm{cls}}z^* + b$ produces class predictions. At equilibrium, $G(z^*, A){=}z^*{-}F(z^*, A){=}0$ defines z^* as an implicit function of A, with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p}\bigg|_{z^*}. \tag{2}$$

Equation (2) is the lever AEGIS pulls: it gives, in closed form, the first-order change in the equilibrium z^* under any parameter p, with the resolvent $(I - J_z)^{-1}$ amplifying a perturbation by at most $1/(1 - \kappa)$ through its convergent Neumann series. The pseudospectral index $\eta{=}\left\|(I - J_z)^{-1}\right\|_2 (1 - \rho) \geq 1$ [26] measures the gap to the spectral-radius bound, equalling 1 for normal J_z. The IFT has been used for hyperparameter optimization [27] and implicit differentiation [19], but not for adversarial structural analysis.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \left\|\delta\hat{A}\right\|_F \leq \varepsilon, \tag{3}$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A}{=}\delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij}{=}0$ for $(i, j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge-deletion / weight-perturbation; insertion attacks (Nettack-class [1], [4]) require enlarging the basis to $\bar{E} \supseteq E$, a natural follow-up that preserves the S_c construction. Discrete deletions connect to the continuous model through Proposition 3. Unlike input-feature perturbation [28], [29], structural perturbation changes the operator itself: z^* depends on A through both $J_z = \mathrm{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F/\partial\mathrm{vec}(A)$, motivating S_c in section III.

III. Adversarial Equilibrium Theory

Building on the threat model of section II, we derive the analytical machinery behind the three diagnostics promised in section I: per-edge rankings, the SVD-optimal direction, and per-node radii. The constrained sensitivity matrix S_c is the unified object from which all three follow.

The governing picture is a phase transition in the perturbation budget ε. While ε stays small, the perturbed model remains contractive and its prediction moves smoothly and boundedly with the graph; as ε approaches a critical budget $\varepsilon_{\mathrm{crit}}$, the equilibrium grows arbitrarily sensitive; past $\varepsilon_{\mathrm{crit}}$, contraction is no longer guaranteed and the prediction can break. Theorem 1 makes this precise and puts $\varepsilon_{\mathrm{crit}}$ in closed form.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\mathrm{proj}})$ be an IGNN-class operator satisfying:*

(A1) *ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT [30] (proof below).*

(A2) *$\|W\|_2 \leq c$ via spectral normalization.*

(A3) *$\|J_z\|_2 \leq \kappa < 1$ verified post-training (κ=0.14–0.59 across all benchmarks in our suite; section V-E). (A2) does not imply (A3) automatically; we report κ alongside $\varepsilon_{\mathrm{crit}}$ so practitioners can audit, and* AEGIS *continues to return S_c rankings and SVD-optimal attacks if (A3) is not satisfied, with the formal guarantees below then scoped to the contractive regime.*

Define the critical budget

$$\varepsilon_{\mathrm{crit}} = \frac{1 - \kappa}{\|W\|_2}. \tag{4}$$

Then structural perturbations $\delta\hat{A}$ with $\left\|\delta\hat{A}\right\|_F = \varepsilon$ exhibit three regimes:

(a) Subcritical *($\varepsilon < \varepsilon_{\mathrm{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \tag{5}$$

where $S = (I - J_z)^{-1}J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\mathrm{op}}/(1 - \kappa)$.

(b) Critical *($\varepsilon \to \varepsilon_{\mathrm{crit}}$): The resolvent obeys the unconditional eigenvalue bound $\left\|(I{-}J'_z)^{-1}\right\|_2 \geq 1/\min_i |1{-}\lambda_i(J'_z)|$, so it blows up precisely when an eigenvalue of J'_z approaches 1: a vanishing spectral gap to 1, not $\|J'_z\|_2 {\to} 1$ on its own. The rate is $\Omega(1/(\varepsilon_{\mathrm{crit}}{-}\varepsilon))$ when J'_z is normal with a dominant real-positive eigenvalue (the Perron mode of symmetric $\hat{A}, W$); in general $\varepsilon_{\mathrm{crit}}$ is the norm-based contraction radius that* lower-bounds *the eigenvalue divergence threshold, the slack reflecting the nonnormality of J'_z (bounded by η, Observation 1; $\eta{\leq}2.47$ for general ReLU, Remark 1).*

(c) Supercritical *($\varepsilon \geq \varepsilon_{\mathrm{crit}}$): The sufficient contraction certificate no longer applies ($\|J'_z\|_2$ may exceed 1) and the part-(a) first-order guarantees do not apply; the certificate is reported as a sufficient safety boundary, validated empirically by the $\kappa_{\max}$ sweep of section V-D (trained IGNNs hold with 2–4× margin).*

Proof. **(a).** Under (A1), F_θ is piecewise-affine on each ReLU linear region; the activation pattern at z^* is generically stable for sufficiently small $\left\|\delta\hat{A}\right\|$ (the boundary set $\{\hat{A} : z^*(\hat{A})$ lies on a region face$\}$ has Lebesgue measure zero), so the conservative-gradient calculus of Bolte–Pauwels [30] yields a conservative IFT on each region. With $J_z = \mathrm{diag}(\phi')(\hat{A} \otimes W)$ and $\kappa {\leq} \left\|\hat{A}\right\|_2 \|W\|_2$, applying IFT to $G(z, \hat{A}){=}z{-}F(z, \hat{A}){=}0$ at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1}J_A \cdot \mathrm{vec}(\delta\hat{A}) + O(\left\|\delta\hat{A}\right\|^2), \tag{6}$$

with $J_A = \partial F/\partial\mathrm{vec}(A)$ and Neumann $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1 - \kappa)$ yielding (5). For contractivity preservation, $\|J'_z\|_2 \leq (\left\|\hat{A}\right\|_2 + \varepsilon) \|W\|_2$, so $\varepsilon < \varepsilon_{\mathrm{crit}}$ guarantees $\|J'_z\|_2 < 1$.

(b)–(c). For any M, $(I{-}M)^{-1}$ has eigenvalues $(1{-}\lambda_i(M))^{-1}$, hence $\left\|(I{-}M)^{-1}\right\|_2 \geq \rho((I{-}M)^{-1}) = 1/\min_i |1{-}\lambda_i(M)|$ *unconditionally*. Along the worst-case direction $\|J'_z\|_2 \leq (\left\|\hat{A}\right\|_2 {+} \varepsilon) \|W\|_2$, so $1{-}\|J'_z\|_2 \geq \|W\|_2 (\varepsilon_{\mathrm{crit}}{-}\varepsilon)$ and, since $\rho {\leq} \|\cdot\|_2$, $\varepsilon {<} \varepsilon_{\mathrm{crit}}$

certifies $\|J_z'\|_2 < 1$. The certificate is one-sided: $\|J_z'\|_2 \to 1$ need not drive $\min_i |1-\lambda_i| \to 0$ (e.g. $M = \mathrm{diag}(-s, 0)$ has $\|M\|_2 = s$ yet $\left\|(I-M)^{-1}\right\|_2 = 1$). When J_z' is normal with dominant real-positive eigenvalue $\lambda^\star = \|J_z'\|_2$ (the Perron mode for symmetric $\hat{A}, W$), $\min_i |1-\lambda_i| = 1-\lambda^\star = \|W\|_2 (\varepsilon_{\mathrm{crit}} - \varepsilon)$ yields the $\Omega(1/(\varepsilon_{\mathrm{crit}} - \varepsilon))$ rate; a negative or complex dominant eigenvalue keeps $\min_i |1-\lambda_i|$ away from 0 (the same $\mathrm{diag}(-s, 0)$), so $\varepsilon_{\mathrm{crit}}$ only lower-bounds the threshold, slack controlled by η (Observation 1). For $\varepsilon \geq \varepsilon_{\mathrm{crit}}$ Banach contraction fails and the iteration may converge elsewhere, oscillate, or diverge. □

$\varepsilon_{\mathrm{crit}}$ is a closed-form sufficient bound that S_c tightens 7–14× over the unconstrained $\sqrt{r}$ bound (section V-A). The Neumann bound uses κ, not $\rho(J_z)$; the gap is the pseudospectral index η, bounded by W alone:

Observation 1 (Graph-Independent Nonnormality Bound, all-active case)**.** *For $J_z = \mathrm{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$ and all activations positive ($\phi' \equiv 1$): $\eta \leq \kappa(V_W)$ where $W = V_W D_W V_W^{-1}$ and $\kappa(V_W) = \|V_W\|_2 \left\|V_W^{-1}\right\|_2$, independent of $\hat{A}$.*

Proof. $J_z = \hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ [31]; symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}}) = 1$, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\left\|(I-M)^{-1}\right\|_2 \leq \kappa(V)/(1-\rho(M))$ yields $\eta \leq \kappa(V_W)$. □

Remark 1 (Empirical extension to general ReLU patterns)**.** *The proof above requires $\phi' \equiv 1$; for general ReLU patterns, $\eta \in [1.19, 2.47]$ on our suite (section V-E), tracking $\kappa(V_W)$ closely. Graph structure does not amplify nonnormality; the moderate η traces to spectral-norm regularisation of W.*

With the resolvent's amplification characterised, we now read the three diagnostics off S_c, beginning with the single most damaging perturbation direction.

Proposition 1 (Maximally Sensitive First-Order Structural Perturbation)**.** *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \mathrm{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S. The maximum first-order equilibrium shift is $\varepsilon \cdot \sigma_1(S)$.*

By the variational characterisation of σ_1, the direction (Proposition 1) is optimal for hidden-state damage within the pseudospectral envelope of Observation 1 and Remark 1, and empirically aligns with classification damage (section V-B). The *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd \times |E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}$ paired with edge basis $b_k = (e_i e_j^\top + e_j e_i^\top)/\sqrt{2}$ ($\|b_k\|_F = 1$, so $\sigma_1(S_c)$ is consistent with $\left\|\delta\hat{A}\right\|_F = \varepsilon$). The $N^2 \to |E|$ projection P_c is the standard duplication-matrix reduction [32] on the edge-supported symmetric subspace; the operational contribution is the matrix-free $S_c v$ in algorithm 1. The first-order envelope identifies the dangerous direction rather than a global safety margin (section V-A).

Proposition 2 (Per-Node First-Order Sensitivity Radius)**.** *For a linear head $f(z) = Wz + b$ and a node v classified as y_v with positive margins $m_v^{(c)} = f_{y_v}(z_v^*) - f_c(z_v^*) > 0$ to every competitor $c \neq y_v$, the first-order sensitivity radius*

$$r_v = \min_{c \neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c)\, S_v\|_2} \tag{7}$$

preserves the classification of v for any $\left\|\delta\hat{A}\right\|_F < r_v$, where S_v are the block-rows of S at node v; substituting $S_{c,v}$ gives the tighter constrained-threat radius $r_v^{(c)} \geq r_v$. Minimizing over all *competitors is the distance to the nearest first-order boundary, so the bound holds even when the runner-up changes under perturbation; the cheaper runner-up surrogate $\hat{r}_v = m_v/(\|W_{y_v} - W_{c^*}\|_2 \|S_v\|_2)$ ($c^* = \arg\max_{c \neq y_v} f_c$) can be optimistic when a non-runner-up class is more aligned.*

Proof sketch. By Thm. 1(a), $\Delta z_v^* \approx S_v \,\mathrm{vec}(\delta A)$, so for each competitor $|\Delta(f_{y_v} - f_c)| \leq \|(W_{y_v} - W_c)S_v\|_2 \|\delta A\|_F$ by Cauchy–Schwarz; the prediction survives while every margin stays positive, i.e. $\|\delta A\|_F < m_v^{(c)} / \|(W_{y_v} - W_c)S_v\|_2$ for all c, whose minimum is r_v. Intuitively, r_v is the smallest margin divided by two amplification factors (decision-boundary sensitivity to hidden-state change, and equilibrium sensitivity to structure); high-margin, low-sensitivity nodes get large radii.

Remark 2 (Sensitivity threshold semantics)**.** *r_v is a first-order sensitivity threshold, not a probabilistic certificate: it indicates the perturbation scale at which the linearization predicts a classification change for node v, and is locally tight for small ε but can be violated at larger magnitudes where higher-order terms dominate. The two approaches are complementary: randomized smoothing [7] certifies which nodes are guaranteed safe, while* AEGIS *differentiates which edges and nodes are most sensitive and in which direction (section V-B). Empirically, breach rate is 0% at $\varepsilon = 0.01$ and rises with ε as expected for first-order approximations (section V-B).*

A. Continuous-to-Discrete Transfer

Proposition 3 (Continuous-to-Discrete Ranking Transfer)**.** *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i,j) \in E$, $v_k = \|[S_c]_{:,k}\|_2$ the continuous vulnerability score, and $d_k = \|z^*(A) - z^*(A \setminus k)\|_2$ the discrete damage from fixed-normalization removal of edge k. Assume all single-edge removals are subcritical: $\sqrt{2}\max_k w_k < \varepsilon_{\mathrm{crit}}$.*
(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \tag{8}$$

where $L_J \leq \|W\|_2^2 \|z^*\|$ is the second-order curvature constant of $z^*(A)$, finite at the fixed point.
(b) Ranking preservation. *For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever*

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J/(1-\kappa)^2}. \tag{9}$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1 \neq k_2}(v_{k_1} - v_{k_2}) \cdot (1-\kappa)^2 / L_J$.

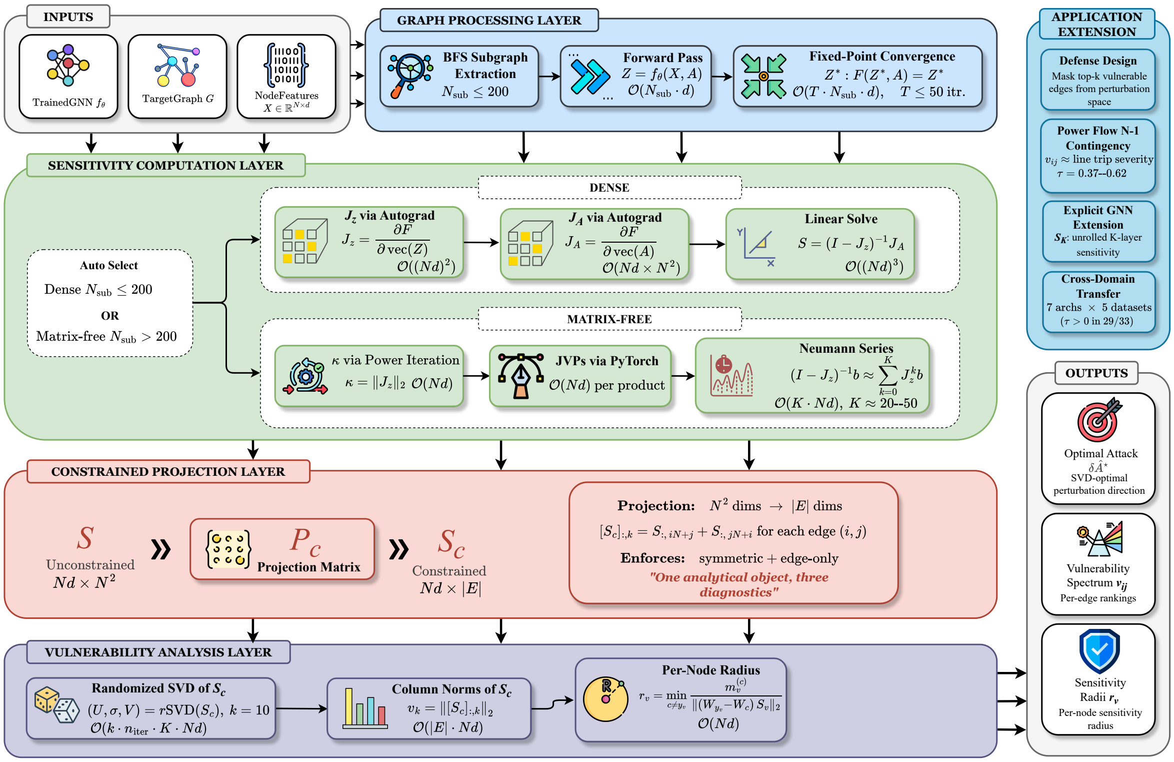


Fig. 1: The four-stage AEGIS pipeline: (1) graph processing (BFS subgraph or full-graph forward / fixed-point pass); (2) sensitivity computation (dense Jacobian for $N \le 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \to |E|$ via P_c; (4) vulnerability analysis (randomised SVD, S_c column norms, per-node radii). The matrix-free path queries S_c via JVPs/VJPs without materialising it, scaling to N=7,650.

Proof. **(a).** We model deletion as *fixed-normalization* masking, $[\delta\hat{A}]_{ij}=[\delta\hat{A}]_{ji}=-w_k$ with the degree matrix D held fixed, matching the continuous edge-weight perturbation, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. (recompute-normalization adds an $O(d_i^{-1})$ incident-edge rescaling, negligible off low-degree nodes; agreement in section V-A.) For the remainder, the IFT gives $\partial z^*/\partial \operatorname{vec}(A)=(I-J_z)^{-1}J_A$, and the dominant curvature term $(I-J_z)^{-1}(\partial J_z/\partial \operatorname{vec}(A))(I-J_z)^{-1}J_A$ carries two resolvents $(1-\kappa)^{-1}$, two factors $\|W\|_2$, and the equilibrium norm $\|z^*\|$ (since $\partial J_z/\partial \operatorname{vec}(A) \propto W$ and $J_A \propto z^* W^\top$), so $L_J \le \|W\|_2^2 \|z^*\|$, finite since $\|z^*\| \le \|X_{\text{proj}}\| / (1- \left\|\hat{A}\right\|_2 \|W\|_2)$ (from $\|\phi(u)\| \le \|u\|$ at the fixed point and the well-posedness contraction $\left\|\hat{A}\right\|_2 \|W\|_2 < 1$ of section II, which implies (A3) since $\kappa \le \left\|\hat{A}\right\|_2 \|W\|_2$), giving $\left\|\partial^2 z^*/\partial \operatorname{vec}(A)^2\right\|_2 \le (1-\kappa)^{-2} L_J$. The path meets finitely many ReLU regions (simultaneous crossings non-generic, measure zero); summing the second-order remainder over these regions, with the conservative IFT [30] on each, gives $|R_k| \le L_J w_k^2/(2(1-\kappa)^2)$. **(b).** From (a), $d_{k_1} - d_{k_2} \ge w_{k_1}v_{k_1} - w_{k_2}v_{k_2} - C(w_{k_1}^2 + w_{k_2}^2)$ with $C = L_J/(2(1-\kappa)^2)$; under $v_{k_1} > v_{k_2}$, $w_{k_1} \ge w_{k_2}$, the first term is positive and (9) ensures the first-order gap dominates. □

Empirically $|R_k|$ is 2–10× smaller than $L_J w_k^2$, and the constrained projection S_c tightens the bound from 0.31 to 1.00 (section V-A). The edge-weighted score $w_k v_k$ predicted by (a) reaches $\tau = +\mathbf{0.996}$ against brute-force N-1 on full-graph Amazon Photo (10 seeds, section V-F), corroborating the first-order bridge at scale.

B. Generalization to Explicit GNNs

Proposition 4 (Structural Sensitivity for K-Layer GNNs)**.** *Let $Z_K = g_K \circ \cdots \circ g_1(X, A)$ be a K-layer GNN where each layer g_l is differentiable with respect to both its input and A. Define the* unrolled sensitivity matrix

$$S_K = \frac{\partial \operatorname{vec}(Z_K)}{\partial \operatorname{vec}(A)} = \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} J_z^{(k)} \right) J_A^{(l)}, \tag{10}$$

where $J_z^{(k)} = \partial g_k/\partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l/\partial \operatorname{vec}(A)$. Then:
***(a)** The first-order shift bound $\|\Delta Z_K\|_F \le \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \le \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} \left\|J_z^{(k)}\right\|_2 \right) \left\|J_A^{(l)}\right\|_2. \tag{11}$$

***(b)** The constrained construction S_c and per-node radius r_v (Proposition 2) apply identically with S_K in place of S.*

For weight-tied models, $\sigma_1(S_K) \le \|J_A\|_2 (1-\kappa^K)/(1-\kappa)$ converges to the IGNN bound as $K \to \infty$. Explicit GNNs inherit the full computational pipeline with empirical tightness 0.99–1.02 across the architecture suite (section V-F); the closed-form $\varepsilon_{\text{crit}}$ is available only for the contractive subclass.

Algorithm 1 AEGIS matrix-free vulnerability analysis. S_c is queried but never formed; the dense fallback ($N \le 200$) builds S_c and runs an exact SVD.

Require: F, fixed point z^*, $\hat{A}$, edge set E, budget ε, rank k, Neumann tolerance tol; (opt.) head W, margins $m_v^{(c)}$.
Ensure: $\sigma_1(S_c)$, $\delta\hat{A}^\star$, $\{v_{ij}\}$, $\{r_v\}$, $\varepsilon_{\text{crit}}$.

$\kappa \leftarrow \|J_z\|_2$; $\quad K \leftarrow \lceil \log(1/\text{tol})/\log(1/\kappa) \rceil$
$S_c\, v := \left(\sum_{j=0}^{K} J_z^j\right) J_A\, P_c\, v$ {matrix-free}
$(\sigma_1, v_1) \leftarrow \text{rSVD}(S_c, k)$ [33]
$\delta\hat{A}^\star \leftarrow \varepsilon\, \text{sym}(P_c v_1) / \|\text{sym}(P_c v_1)\|_2$
$v_{ij} \leftarrow \|S_c\, e_{ij}\|_2$ for each $(i,j) \in E$
$r_v \leftarrow \min_{c \neq y_v} m_v^{(c)} / \|(W_{y_v} - W_c)\, S_{c,v}\|_2$ {head only}
$\varepsilon_{\text{crit}} \leftarrow (1-\kappa)/\|W\|_2$ {Thm. 1}
return $(\sigma_1,\ \delta\hat{A}^\star,\ \{v_{ij}\},\ \{r_v\},\ \varepsilon_{\text{crit}})$

IV. AEGIS CONSTRUCTION

The pipeline makes the theory of section III practical: it queries S_c matrix-free, never materialising $S \in \mathbb{R}^{Nd \times N^2}$ or forming the resolvent. One S_c computation returns the per-edge spectrum $\{v_{ij}\}$, the SVD-optimal direction $\delta\hat{A}^\star$, and per-node radii $\{r_v\}$, plus $\varepsilon_{\text{crit}}$ and the regime characterisation of Theorem 1 for IGNN-class operators. A dense path ($N \le 200$) computes an exact SVD; a matrix-free path ($N > 200$, validated to N=7,650) uses JVP/VJP queries and randomized SVD. Figure 1 illustrates the four stages; algorithm 1 summarises the matrix-free routing.

Matrix-free operator. The key to scaling is to never build S_c: we only apply it to vectors, which needs $O(Nd)$ memory rather than the $O((Nd)^2)$ a dense S_c would cost. Concretely, $S_c v = (I - J_z)^{-1} J_A P_c v$ is evaluated via the truncated Neumann series $\sum_{k=0}^{K} J_z^k b$ with depth K adaptive in κ (early stop at $\|J_z^k b\| < 10^{-6}\|b\|$); each Neumann term and $J_A\, \text{vec}(\delta A)$ are forward-mode JVPs in $O(Nd)$ time. P_c is applied implicitly, so neither S_c nor S is formed; the dense path ($N \le 200$) materialises J_z, J_A and applies a direct linear solve. The rSVD [33] (k=p=10, n_{iter}=2) returns (u_1, σ_1, v_1) for the direction $\delta\hat{A}^\star = \varepsilon \cdot \text{sym}(P_c v_1)$ (Proposition 1), where $\text{sym}(M) = (M + M^\top)/2$ keeps $\delta\hat{A}^\star$ symmetric (section II); column norms give $\{v_{ij}\}$, and per-node radii follow Proposition 2 via the node-restricted RMATVEC. The well-separated singular spectrum (gap $(\sigma_1 - \sigma_2)/\sigma_1$=0.39–0.50; fig. 2) is why one rSVD query matches iterative attackers (section V-B).

Cost. Each $S_c v$ is $O(K \cdot Nd)$ time and $O(Nd)$ memory ($K \in [20, 50]$ for $\kappa < 0.8$); removing the $O((Nd)^3)$ solve lifts the previous $N \approx 300$ ceiling (section V-D).

V. EXPERIMENTS

Our evaluation tests four claims in turn: the contractivity assumptions hold on trained models (section V-A); the one-query SVD direction matches iterative attackers (section V-B); the continuous S_c ranking transfers to discrete edge removal across architectures (section V-F); and the same object screens power-grid contingencies (section VI).

Setup. We evaluate AEGIS on 10 datasets across 5 domains with 10 random seeds each (PyTorch [34]); reported

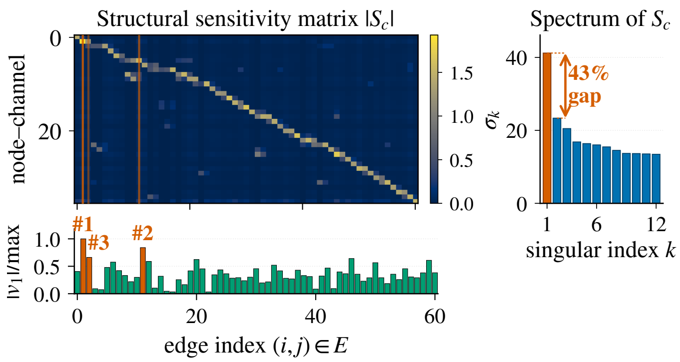


Fig. 2: $|S_c|$ on a 50-node Cora ego-graph (IGNN, N=50, d=16, $|E|$=61, seed 0). Vermilion lines: top-3 edges by $|v_1|$ (SVD-optimal direction, bar strip). Sidebar: singular spectrum, σ_1=41.2, 43% gap to σ_2 (cross-benchmark $(\sigma_1 - \sigma_2)/\sigma_1 \in [0.39, 0.50]$; section V-B).

TABLE I: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$ is the theoretical critical budget. AtkAdv: AEGIS damage / random damage. Cov%: fraction of subgraph nodes certified first-order-safe at budget ε=0.05 (i.e. corrected radius r_v>0.05).

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	$77.5_{\pm 1.7}$	$.33_{\pm .14}$	$.66_{\pm .15}$	$3.6_{\pm .6}$	$83_{\pm 10}$
Citeseer	$66.0_{\pm 0.7}$	$.59_{\pm .02}$	$.41_{\pm .02}$	$4.1_{\pm .5}$	$94_{\pm 5}$
Pubmed	$78.9_{\pm 0.7}$	$.41_{\pm .11}$	$.59_{\pm .11}$	$3.2_{\pm .4}$	$76_{\pm 18}$
Amazon	$94.8_{\pm 0.3}$	$.14_{\pm .02}$	$.86_{\pm .02}$	$3.8_{\pm .2}$	$86_{\pm 8}$
WikiCS	$77.9_{\pm 0.4}$	$.34_{\pm .04}$	$.66_{\pm .04}$	$3.8_{\pm .4}$	$78_{\pm 4}$

$\pm$ are standard deviations over seeds. IGNN [22] uses $F(Z) = \text{ReLU}(\hat{A}ZW^\top + U(X))$ with $W \in \mathbb{R}^{64 \times 64}$ spectral-normalised [23] and Adam (lr 0.01); ContractiveGCN-PF mirrors this for power flow. The spectral-norm constraint costs ∼6% Cora accuracy. Datasets: Cora, Citeseer, Pubmed [35], Amazon Photo [36], WikiCS [37], Amazon Fraud [38], IEEE 14/30/57/118 [39], [40]. Baselines: smoothing (Gaussian σ=0.01), Mettack (Meta-Self GCN surrogate), LODF as $\text{PTDF}_{ij}/(1 - \text{PTDF}_{jj})$ from the DC B-matrix [41, Ch. 11]. Coverage is not uniform across studies: all seven architectures appear only in table IV and fig. 7 (39 cells), while the other classification studies use IGNN and the power-flow study uses the four IEEE cases.

A. Cross-Domain Adversarial Analysis

Two things must hold before the rest of the evaluation means anything: the trained models should sit in the contractive regime the theory assumes, and AEGIS's perturbations should be genuinely more damaging than chance. Table I confirms both, verifying (A3) on every benchmark (κ<1) and reporting the per-dataset closed-form $\varepsilon_{\text{crit}}$ alongside an attack advantage of 3.2–4.1× over random perturbations. Unless otherwise stated, IGNN experiments in section V-A–section V-E use 50-node BFS ego-subgraphs centred on the highest-degree node and report *local vulnerability*; graph-level assessment uses the matrix-free pipeline of section V-D. All bounds use the operator-norm $\kappa = \|J_z\|_2$.

AEGIS's shift bounds are first-order, so a natural question is how far they hold as the budget grows. Predicted and actual equilibrium shifts agree to within 1% at the small budgets that

TABLE II: Equilibrium damage at ε=0.10 across attack methods (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS. Cls-PGD: classification-loss PGD. Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	SVD	Cls-PGD	Shift-PGD	Random
Cora	$3.70_{\pm.79}$	$2.51_{\pm.48}$	$3.05_{\pm.70}$	$0.97_{\pm.25}$
Citeseer	$4.63_{\pm.71}$	$2.97_{\pm.42}$	$3.35_{\pm.57}$	$1.01_{\pm.15}$
WikiCS	$2.10_{\pm.22}$	$0.67_{\pm.11}$	$1.93_{\pm.20}$	$0.53_{\pm.09}$

matter for early warning, and loosen to at most 39% on the citation graphs (within 7% on WikiCS and Amazon) even at ε=0.20. This envelope ratio is measured on the equilibrium shift $\|\Delta z^*\|_F$, distinct from the prediction-flip (breach) rate of fig. 4.

Heuristic and surrogate baselines. AEGIS beats degree-proportional, edge-betweenness, and spectral rankings at ε=0.01 (+6–148% AtkAdv, Wilcoxon p<0.001) and inflicts 3–10× more equilibrium damage than Mettack [4] at the early-warning regime $k \in \{1, \dots, 5\}$ ($\mathbf{149/150}$ paired wins, p<10^{-43}); GR-BCD/PR-BCD [5] and the discrete-removal curves are in table III and fig. 3.

B. Four-Quadrant Attack Comparison

Radius vs. smoothing. Randomized smoothing [7] gives constant-per-node base radii (0.123 vs. AEGIS's 0.078 at σ=0.05), and localized smoothing [9] needs 10^3–10^4 MC/node; both lack edge structure. AEGIS radii are instead structurally informative (r_v≈0.10 in dense regions, 0.01 at boundaries), a complementary point on the certificate-versus-diagnostic frontier.

Table II evaluates AEGIS across the four quadrants of (gradient-based, gradient-free) × (equilibrium-shift, classification-loss). The SVD direction maximises first-order equilibrium shift *by construction* (Proposition 1); what matters is non-circularity and cost. A transfer direction from a separately trained surrogate (zero shared gradients) recovers 99% of the one-query damage (cos = 0.99), versus 44±4% for a 512-query black-box search: model-intrinsic, not a gradient artifact. Per query it delivers 74–156× the damage of a 50-step PGD [42]; classification-loss PGD inflicts 15–70% less at 50× the cost, IFT-gradient PGD (solver validation, not independent) 72–92%, and prediction *flips* stay 0–1.8% for all methods. AEGIS is a one-query *diagnostic*, not a peak attacker: budget-heavy attackers (GR-BCD, table III) exceed it at larger budgets.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20_{\pm 0.05}$ vs. AEGIS's $+0.32_{\pm 0.04}$) and stay near zero elsewhere; AEGIS achieves 2–6× higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces.

Per-edge finite differences reproduce the S_c column-norm ranking (τ=0.999); iterative S_c recomputation closes ≤ 3 pp of the static-vs-greedy gap at $k\times$ compute, so the static ranking is the headline.

Breach rates. Every breached node satisfies $\varepsilon > r_v$ across all datasets and budgets, validating r_v as a reliable first-order screening threshold (Remark 2). Figure 4 reports prediction-flip fractions across ε (bands are 95% CIs): Citeseer and WikiCS stay below 2% throughout; Cora reaches 7.6% (CI $[2.2, 13.0]$) at ε=0.20; Pubmed is the right-skewed outlier (median 7.8%, mean 10.3% at ε=0.10; 27.4%, CI $[12.3, 42.5]$, at ε=0.20).

TABLE III: AEGIS vs. external baselines (10 seeds). GR-BCD [5] is the BCD-family scalable structural attacker (PR-BCD is the same paper's larger-budget variant, see text); AGNNCert [11] is a sound IBP certifier (semantics differ from AEGIS r_v); LODF [41] is the industry DC-PF screener.

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD [5]	Pubmed k=10	0.356	0.350	$\mathbf{+0.69}$
GR-BCD [5]	Cora k=5	0.643	1.207	+0.16
AGNNCert† [11]	Cora med. r_v	0.163	1.414	+0.08
LODF [41]	case118 P@10	0.81	≤ 0.20	+0.14

†AGNNCert is a *sound* IBP certificate; AEGIS r_v is a first-order sensitivity threshold (Remark 2). Per-dataset mean tightness ratio $r_{\text{cert}}/r_v \in [4.4, 15.0]$ (per-cell across 30 seeds $[2.1, 39.0]\times$); per-seed Kendall $\tau \in [-0.11, 0.24]$. **Decision rule:** if AGNNCert certifies node v safe at budget ρ, trust it; if $r_v < \rho$ and AGNNCert does *not* certify, v is first-order unsafe and warrants attention. The methods are complementary in this prescribed sense, not interchangeable.

C. Comparison with Prior Structural Baselines

Table III reports head-to-head numbers against GR-BCD and PR-BCD [5], AGNNCert [11], and LODF [41]. AEGIS is a label-free one-query proxy; the question is how much of each gold-standard signal it retains.

GR-BCD [5] aligns with S_c on Pubmed (τ=+0.69) but decorrelates on Cora (τ=+0.16); as a label-free one-pass diagnostic rather than a budget-optimal attacker, AEGIS is competitive rather than dominant under direct attack damage (table III). AGNNCert [11] is complementary: a sound IBP certificate, against which AEGIS's first-order radii run 4.4–15.0× tighter while also exposing the dominant direction (decision rule: table III). Power-grid LODF/PI comparisons are in section VI.

D. Phase Transition and Scalability

Phase transition. Figure 6 sweeps $\kappa_{\max} \in [0.30, 0.99]$ on Cora (10 seeds). Even when the spectral-normalisation cap is pushed to 0.99, driving the theoretical $\varepsilon_{\text{crit}}$ down 230×, the trained ReLU pattern keeps $\rho(J_z) \leq 0.42$, so the resolvent grows only $\mathbf{1.17 \to 1.80}$. This *validates* $\varepsilon_{\text{crit}}$ as a closed-form sufficient safety boundary, one that holds with $\mathbf{2-4}\times$ empirical margin across the sweep. Training converges throughout, and the matrix-free pipeline reaches its 50-step fixed-point budget at $\kappa_{\max} \geq 0.85$.

Scalability and matrix-free self-consistency. Figure 5 compares dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at N=200 (per-edge τ=0.999), and the analytical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite (consistent with the trained-IGNN range $\kappa \in [0.14, 0.59]$ of table I). For larger N the rSVD error [33] is bounded by the spectral gap; Pubmed (N=19,717) exceeds 24 GB.

E. Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N \leq 100$ and

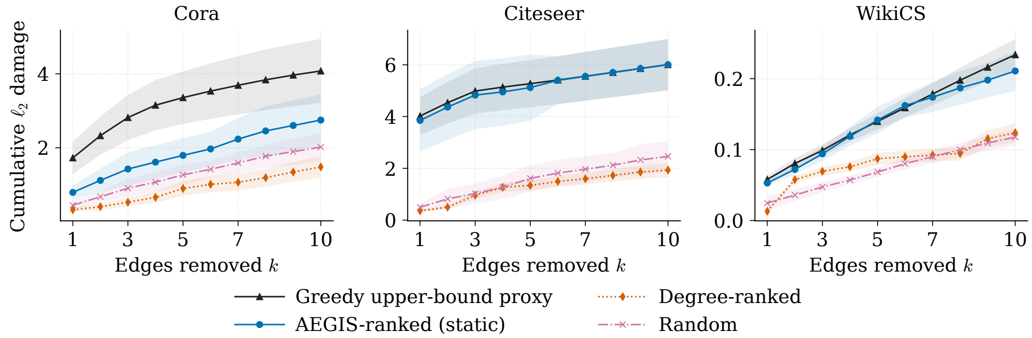


Fig. 3: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds, ± 1 sd). AEGIS *matches* the ground-truth-aware Greedy attacker on Citeseer/WikiCS and recovers **54–67**% of Greedy's damage on Cora at k=5–10 *without any access to discrete ground truth*; degree-ranked stays below random on Cora at every k.

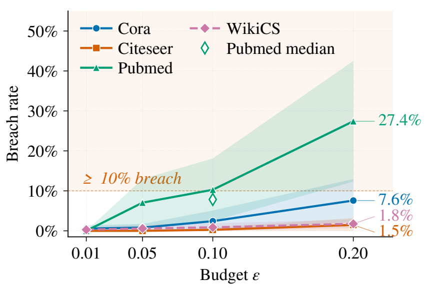


Fig. 4: Breach rate (% of nodes flipped under S_c-optimal perturbation) vs. ε (10 seeds; mean, 95% CI bands).

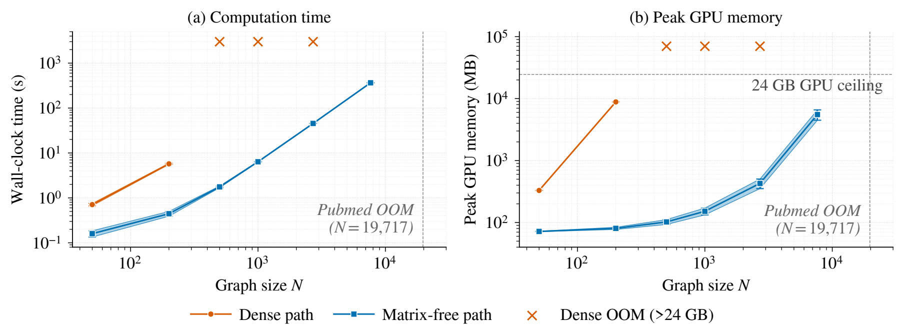


Fig. 5: Dense vs. matrix-free AEGIS pipeline (RTX 4090). Dense scales as $O((Nd)^3)$, exceeding 24 GB beyond N=200; matrix-free is roughly $O(N \log N)$ in time, sub-linear in memory, reaching Amazon Photo (N=7,650) at 365 s/5.5 GB.

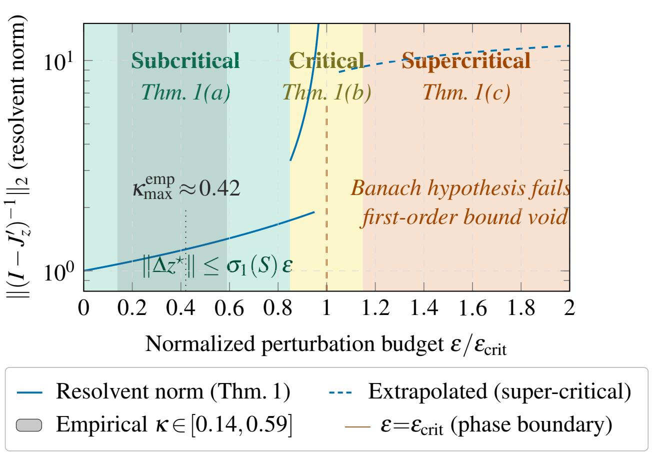


Fig. 6: Three-regime characterisation of Theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1-\kappa)$ divergence at $\varepsilon \to \varepsilon_{\text{crit}}$.

degrades to 1.031 at N=200; we use N=50 as default (66× faster). Subgraph-vs-full-graph agreement is high on dense grids (case14 τ=0.80–0.86).

Full-graph scale. A 50-node BFS covers only $\sim$1.8% of a citation graph's edges (Cora τ=0.16), so at this scale we run AEGIS on the full graph. There its edge advantage over degree-proportional ranking *amplifies* to **9.82**× (Citeseer) and **3.25**× (Cora) at k=10, against $\approx$1.1× on subgraphs.

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$ traces the accuracy–robustness frontier (r_v=0.147@72.1% vs. 0.089@80.6%).

Defense-informed edge protection: gains survive adaptive attack. Masking the top-k edges by v_{ij} (Cora IGNN, N=50, 10 seeds) cuts $\sigma_1(S_c)$-damage by **42**±**8**% at k=5 and **61**±**7**% at k=10, vs. 11±6%/18±9% for random masking (paired Wilcoxon $p<0.002$). An *adaptive* attacker that recomputes S_c on the masked graph matches the non-adaptive attacker within 0.1 pp (10 seeds, ε=0.10): the 43% spectral gap of S_c (fig. 2) keeps the dominant direction robust, so the defense *does not erode under recomputation*, the failure mode that sober-look critiques [16] flag.

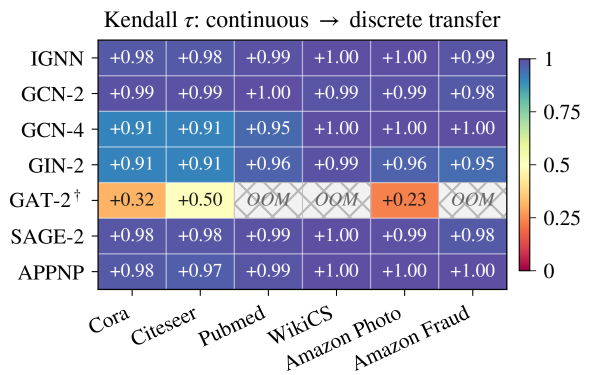
F. Extension to Explicit GNNs

The construction is not tied to implicit models. For K-layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT† [43], [44], [45], [46], [47]) we form S_K=$\partial\text{vec}(Z_K)/\partial\text{vec}(A)$ via finite differences and build S_c from S_K identically (Proposition 4). Table IV reports the per-architecture characterisation on Cora: IGNN carries Theorem 1; the explicit models receive the computational tool without the regime characterisation.

GAT† scope. Standard GAT uses A as a binary mask, so $\partial Z/\partial A_{ij}$=0 on existing edges and S_c is undefined; our GAT† modulates attention by the edge weight ($h_i^\prime = \sum_j \hat{A}_{ij}\alpha_{ij} W h_j$), restoring differentiability and yielding tightness 0.99, AtkAdv

TABLE IV: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ: Kendall vs. brute-force fixed-normalization single-edge removal (Proposition 3). Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	$1.01_{\pm.00}$	$7.6_{\pm0.5}\times$	$+.32_{\pm.04}$	$77.5_{\pm1.7}\%$
GCN-2	$1.00_{\pm.00}$	$3.6_{\pm0.6}\times$	$-.04_{\pm.03}$	$78.9_{\pm0.9}\%$
GCN-4	$1.02_{\pm.00}$	$3.4_{\pm0.4}\times$	$+.49_{\pm.02}$	$78.3_{\pm1.8}\%$
GIN-2	$1.01_{\pm.00}$	$2.6_{\pm0.1}\times$	$+.33_{\pm.03}$	$76.6_{\pm1.9}\%$
GAT†	$1.01_{\pm.01}$	$2.1_{\pm0.1}\times$	$+\mathbf{.54}_{\pm.06}$	$77.8_{\pm1.2}\%$
SAGE-2	$1.00_{\pm.00}$	$1.9_{\pm0.2}\times$	$+.22_{\pm.10}$	$77.5_{\pm1.5}\%$
APPNP	$1.02_{\pm.00}$	$3.4_{\pm0.3}\times$	$+.35_{\pm.01}$	$\mathbf{82.2}_{\pm0.6}\%$



† *GAT-2: 2-layer attention; OOM on Pubmed, WikiCS, Amazon Fraud.*

Fig. 7: Continuous-to-discrete transfer τ (10-seed means, 390 runs; per-cell sd 0.0003–0.041), ranking edges by the edge-weighted $A_{ij}v_{ij}$ of Proposition 3. All 39/39 cells positive (median +0.99); cross-hatch: GPU OOM. GAT-2† is lower under weighting (it transfers better unweighted; see text).

$4.4\times$, $\tau = +0.36$. Binary-mask architectures (hard attention, max/min aggregation, GATv2 [48]-style dynamic-attention masking) fall outside the framework.

Robust backbones. Under the $\sigma_1(W) \le 0.9$ cap, RobustGCN-lite [13] and GNNGuard-lite [14] match or exceed IGNN; the cap is a precondition (without it $\kappa > 1$ voids Theorem 1).

Cross-dataset transfer. We rank edges by the score Proposition 3 predicts, the edge-weighted $A_{ij}v_{ij}$. Figure 7 reports its τ over 6 datasets and 7 architectures (390 runs): *all* 39/39 *cells positive*, median $\tau = +\mathbf{0.99}$ (sign test $p < 10^{-5}$), vs. 34/39 for unweighted v_{ij}. It beats a pure edge-weight baseline by $\Delta\tau \approx +\mathbf{0.25}$, rising to $+\mathbf{0.65}$ on the fraud graph where uniform weights are uninformative, so v_{ij} adds rank information beyond the weight. Weighting matters most at full-graph scale: on Amazon Photo (N=7,650) it reaches $\tau = +\mathbf{0.996} \pm \mathbf{0.002}$ (stratified top-v_{ij} N-1). GAT-2† is the exception: attention reweights edges, so the unweighted v_{ij} transfers better there (+0.47 vs. +0.35).

Proposition 3's sufficient pairwise condition holds for $\mathbf{47}$–$\mathbf{62}\%$ of edge pairs, so the global τ is empirical, not implied. Where unweighted v_{ij} is uninformative (GCN-2's near-uniform 2-hop shifts), the weighted ranking recovers the order via the edge weight (−0.28→+0.99, Citeseer).

VI. Case Study: Power Flow Contingency

Beyond classification, AEGIS extends to AC power-flow contingency analysis. Applied to a learned ContractiveGCN-PF surrogate, the S_c ranking recovers brute-force N-1 critical-line severity on IEEE case14–118 (τ=+0.37 to +0.62, $\mathbf{P@10{=}0.66}$–$\mathbf{0.81}$) without access to the admittance matrix that exact LODF screening requires. The ranking is label-free and one-query: v_{ij} comes purely from the surrogate's learned IFT-resolvent $(I - J_z)^{-1}$, and its agreement with PTDF/LODF [41] is consistent with the topology-driven flow concentration those DC measures encode. Recent GNN-PF work targets fidelity [49], [50], [51], [3] and contingency screening [52]; AEGIS adds a label-free vulnerability-attribution layer over any such surrogate, without contingency labels or retraining.

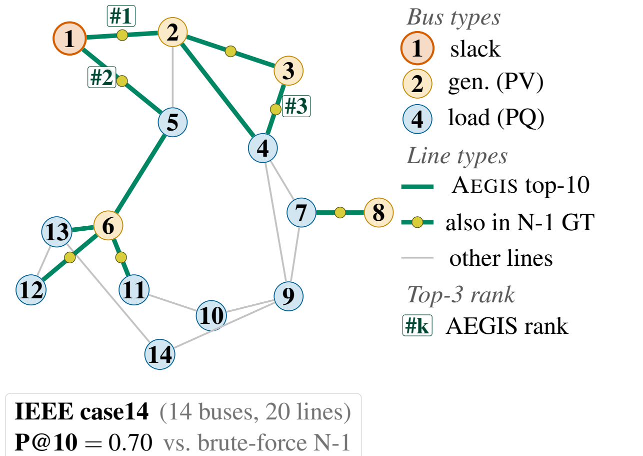


Fig. 8: AEGIS vulnerability ranking on the IEEE 14-bus benchmark (10-seed mean; P@10 = 0.70 for this run, consistent with the 0.74 ± 0.12 in table V). Bus markers encode physical bus type (slack / generator / load); green edges are the top-10 by v_{ij} with rank tags (#1–#3); yellow midpoint dots mark edges that also appear in the brute-force N-1 top-10 (7 of 10). Top-ranked edges concentrate on the high-flow inter-area tie lines (1–2, 1–5, 7–8), recovering the single-contingency severity pattern from one matrix-free SVD pass.

Setup. ContractiveGCN-PF is an IGNN-class operator $F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ with spectral-norm-constrained W, trained on five IEEE test cases [39] with a 2-head readout $(\Delta|V|, \Delta\theta)$. Training data are 2,000 load samples per case (70–130$\%$ of nominal, *uniform load scaling only*, a narrow envelope that does not cover seasonal peaks, dispatch shifts, or renewable ramps) from PandaPower's [40] AC Newton-Raphson solver; inputs are binary adjacency and 5 bus features (slack/PV indicators, P, Q, $|V|$). Ground truth: ℓ_2 voltage-angle deviation per AC contingency. We use full-graph dense analysis on case14–118. LODF's native metric is DC line-flow change; we benchmark its rank-order against AC voltage-angle truth, where the comparison stays informative within LODF's scope.

table V reports the envelope ratio and Kendall τ between S_c rankings and brute-force N-1; fig. 8 visualises case14. The envelope is tight in the PF domain ($\approx$ 1.00), and on case14–118 the ranking correlation τ=+0.37 to +0.62 (P@10=0.66–0.81) identifies 7–8 of the top-10 critical lines. The matrix-free S_c pipeline scales to N=7,650 (fig. 5); richer PF surrogates trained on broader operating envelopes are the path to operational-grade screening.

TABLE V: AEGIS on IEEE power-flow benchmarks (10 seeds). Model quality: per-unit RMSE for $|V|$ and angle θ (std < 0.003). τ: Kendall correlation against brute-force N-1; P@10: top-10 precision.

			Model (p.u.)		AEGIS	
Case	N	$\|E\|$	$\|V\|$	θ	τ	P@10
case14	14	20	.007	.020	$+.42_{\pm.19}$	$.74_{\pm.12}$
case30	30	41	.014	.024	$+.37_{\pm.17}$	$.68_{\pm.06}$
case57	57	78	.033	.059	$+.67_{\pm.09}$	$.66_{\pm.13}$
case118	118	179	.014	.076	$+.62_{\pm.11}$	$.81_{\pm.10}$

Baselines. Against the AC voltage-angle truth, LODF (industry DC screening) attains τ=0.44–0.58 on the credible cases; AEGIS matches or leads it (0.62–0.67 on case57/118, Wilcoxon p<0.01; overlapping on case14/30), and on LODF's fairer thermal-overload target reaches P@10=0.60 (case57), competitive but not dominant. *Binary adjacency beats admittance-weighting* (P@10=0.81 vs. 0.27, case118): a single-line trip removes the line regardless of impedance.[1]

VII. Related Work

We position AEGIS against three threads; each supplies at most one of the three diagnostics AEGIS produces from one S_c object via a single matrix-free pipeline.

Structural attacks. Nettack [1], Mettack [4], GR-BCD/PR-BCD [5] and variants [6] optimise surrogate gradients to flip predictions but provide no analytic maximally sensitive first-order direction and no per-node radius. Sober-look studies [16], [17] flag evaluation pitfalls; we follow their adaptive-attack discipline in section V-E (paired Wilcoxon against random masking; an adaptive-attacker column in defense ablation).

Certified robustness and defense. Convex relaxations [12], smoothing [7], [8], [9], collective certificates [10], and IBP [11] certify per-node robustness but do not identify vulnerability-driving edges. Empirical defenses [13], [15], [14] harden models without surfacing per-edge sensitivity. r_v is a first-order threshold, not a sound certificate (Remark 2); the methods are complementary under the prescribed decision rule of table III.

Implicit networks and influence functions. DEQs [20], IGNN [22], monotone equilibria [54], and well-posedness analyses [28] address *input* sensitivity $\partial z^*/\partial x$; AEGIS targets *structural* sensitivity $\partial Z/\partial A$ via the resolvent $(I - J_z)^{-1} J_A$ and the unrolled Jacobian S_K. The construction descends from the Koh–Liang influence-function lineage [18] and from implicit-differentiation hyperparameter optimisation [27], [19], but those lines apply IFT to feature- or weight-perturbations of a loss; AEGIS applies it to a *structural-parameter* perturbation of the equilibrium state, projects onto the edge subspace via P_c, and obtains the SVD-optimal direction in a single matrix-free pass rather than via iterative reverse-mode. GNN explainers [55] target prediction fidelity (which subgraph rationalises an output), not adversarial vulnerability; classical matrix perturbation theory [31], [26] supplies the pseudospectral toolkit.

[1]Other baselines (case57/118): Ejebe–Wollenberg PI [53] τ=$+0.335/+0.101$ (P@10=0.50/0.30); standalone PTDF without outage correction is anti-correlated/near-zero (τ=$-0.14/+0.06$), confirming outage redistribution is the operative effect. Runtime: LODF <0.13 s, N-1 0.1–2 s, AEGIS 2–23 s.

VIII. Conclusion

AEGIS maps a GNN's *adversarial fault lines*, producing three diagnostics (per-edge rankings, the SVD-optimal attack direction, and per-node first-order sensitivity radii) from one analytical object, the constrained sensitivity matrix S_c. The construction is matrix-free and applies to any GNN with continuous edge-weight-modulated message passing; for contractive implicit models it adds a regime characterisation (Theorem 1) that holds as a sufficient safety boundary.

Limitations. (i) Radii r_v are first-order thresholds, not probabilistic certificates (Remark 2); standard GAT and binary-mask architectures fall outside the framework (section V-F). (ii) Full-graph matrix-free analysis is the recommended graph-level path (section V-F). (iii) Opting into the formal $\varepsilon_{\text{crit}}$ track accepts a $\sim$6% accuracy cost, on par with certified-robustness trade-offs. (iv) Insertion attacks (Nettack-class) are deliberately scoped out; the S_c construction extends once the edge basis is enlarged to $\bar{E} \supseteq E$ (section II).

Disclosure protocol. (i) A 90-day coordinated-notification window before public release, addressed to the maintainers of OGB graph-robustness benchmarks and to the GR-BCD / Mettack / AGNNCert authors. (ii) Attack-direction code (Algorithm 1 steps 3–4) gated behind institutional-affiliation review mediated by an institutional research-ethics office (named in the camera-ready). (iii) The diagnostic-only path (r_v and v_{ij} scores; no SVD reconstruction) released unconditionally, since it cannot directly synthesise a perturbation. Power-grid artefacts use only public IEEE benchmarks [39].

References

[1] D. Zügner, A. Akbarnejad, and S. Günnemann, "Adversarial attacks on neural networks for graph data," in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2018, pp. 2847–2856.

[2] H. Dai, H. Li, T. Tian, X. Huang, L. Wang, J. Zhu, and L. Song, "Adversarial attack on graph structured data," in *International Conference on Machine Learning*, 2018, pp. 1115–1124.

[3] A. M. Nakiganda, C. Weeraddana, N. Popli, and T. van der Hagen, "Graph neural networks for fast contingency analysis of power systems," *arXiv preprint arXiv:2310.04213*, 2023.

[4] D. Zügner and S. Günnemann, "Adversarial attacks on graph neural networks via meta learning," in *International Conference on Learning Representations*, 2019.

[5] S. Geisler, T. Schmidt, H. Sirin, D. Zügner, A. Bojchevski, and S. Günnemann, "Robustness of graph neural networks at scale," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 7637–7649.

[6] H. Wu, C. Wang, Y. Tyshetskiy, A. Docherty, K. Lu, and L. Zhu, "Adversarial examples on graph data: Deep insights into attack and defense," in *International Joint Conference on Artificial Intelligence*, 2019, pp. 4816–4823.

[7] A. Bojchevski, J. Klicpera, and S. Günnemann, "Efficient robustness certificates for discrete data: Sparsity-aware randomized smoothing for graphs, images and more," in *International Conference on Machine Learning*, 2020, pp. 1003–1013.

[8] A. Bojchevski and S. Günnemann, "Certifiable robustness to graph perturbations," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[9] J. Schuchardt, Y. Scholten, A. Bojchevski, and S. Günnemann, "Localized randomized smoothing for GNNs," in *International Conference on Machine Learning*, 2023.

[10] J. Schuchardt, A. Bojchevski, J. Klicpera, and S. Günnemann, "Collective robustness certificates: Exploiting interdependence in graph neural networks," in *International Conference on Learning Representations*, 2021.

[11] Y. Li *et al.*, "AGNNcert: Deterministic certification for graph neural networks against adversarial perturbations," in *International Conference on Learning Representations*, 2025.

[12] D. Zügner and S. Günnemann, "Certifiable robustness and robust training for graph convolutional networks," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 246–256.

[13] D. Zhu, Z. Zhang, P. Cui, and W. Zhu, "Robust graph convolutional networks against adversarial attacks," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 1399–1407.

[14] X. Zhang and M. Zitnik, "GNNGuard: Defending graph neural networks against adversarial attacks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9263–9275.

[15] W. Jin, Y. Ma, X. Liu, X. Tang, S. Wang, and J. Tang, "Graph structure learning for robust graph neural networks," in *Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 2020, pp. 66–74.

[16] F. Mujkanovic, S. Geisler, S. Günnemann, and A. Bojchevski, "Are defenses for graph neural networks robust?" in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 8940–8952.

[17] L. Gosch, S. Geisler, D. Sturm, and S. Günnemann, "Adversarial training for graph neural networks: Pitfalls, solutions, and new directions," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[18] P. W. Koh and P. Liang, "Understanding black-box predictions via influence functions," in *International Conference on Machine Learning*. PMLR, 2017, pp. 1885–1894.

[19] S. Gould, R. Hartley, and D. Campbell, "Deep declarative networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 8, pp. 3988–4004, 2021.

[20] S. Bai, J. Z. Kolter, and V. Koltun, "Deep equilibrium models," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[21] S. Bai, V. Koltun, and J. Z. Kolter, "Multiscale deep equilibrium models," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5238–5250.

[22] F. Gu, H. Chang, W. Zhu, S. Sojoudi, and L. El Ghaoui, "Implicit graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 11 984–11 995.

[23] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, "Spectral normalization for generative adversarial networks," in *International Conference on Learning Representations*, 2018.

[24] S. W. Fung, H. Heaton, Q. Li, D. McKenzie, S. Osher, and W. Yin, "JFB: Jacobian-free backpropagation for implicit networks," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 6, 2022, pp. 6648–6656.

[25] S. Bai, V. Koltun, and J. Z. Kolter, "Stabilizing equilibrium models by Jacobian regularization," in *International Conference on Machine Learning*, 2021, pp. 554–565.

[26] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press, 2005.

[27] J. Lorraine, P. Vicol, and D. Duvenaud, "Optimizing millions of hyperparameters by implicit differentiation," in *International Conference on Artificial Intelligence and Statistics*, 2020, pp. 1540–1552.

[28] L. El Ghaoui, F. Gu, B. Travacca, A. Askari, and A. Tsai, "Implicit deep learning," *SIAM Journal on Mathematics of Data Science*, vol. 3, no. 3, pp. 930–958, 2021.

[29] M. Revay, R. Wang, and I. R. Manchester, "Lipschitz bounded equilibrium networks," in *arXiv preprint arXiv:2010.01732*, 2020.

[30] J. Bolte and E. Pauwels, "Conservative set valued fields, automatic differentiation, stochastic gradient descent and deep learning," *Mathematical Programming*, vol. 188, pp. 19–51, 2021.

[31] G. W. Stewart and J.-g. Sun, *Matrix Perturbation Theory*. Academic Press, 1990.

[32] J. R. Magnus and H. Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, 3rd ed. Wiley, 2019.

[33] N. Halko, P.-G. Martinsson, and J. A. Tropp, "Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions," *SIAM Review*, vol. 53, no. 2, pp. 217–288, 2011.

[34] A. Paszke, S. Gross, F. Massa, A. Lerner, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[35] P. Sen, G. Namata, M. Bilgic, L. Getoor, B. Galligher, and T. Eliassi-Rad, "Collective classification in network data," *AI Magazine*, vol. 29, no. 3, pp. 93–106, 2008.

[36] O. Shchur, M. Mumme, A. Bojchevski, and S. Günnemann, "Pitfalls of graph neural network evaluation," in *Relational Representation Learning Workshop, NeurIPS*, 2018.

[37] P. Mernyei and M. Caron, "Wiki-CS: A Wikipedia-based benchmark for graph neural networks," in *arXiv preprint arXiv:2007.02901*, 2020.

[38] Y. Dou, Z. Liu, L. Sun, Y. Deng, H. Peng, and P. S. Yu, "Enhancing graph neural network-based fraud detectors against camouflaged fraudsters," in *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 2020, pp. 315–324.

[39] IEEE PES, "IEEE PES test feeder working group: Power system test cases," *IEEE Power and Energy Society*, 2024, https://electricgrids.engr.tamu.edu.

[40] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, "pandapower – an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018.

[41] A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*, 3rd ed. John Wiley & Sons, 2014.

[42] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *International Conference on Learning Representations*, 2018.

[43] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *International Conference on Learning Representations*, 2017.

[44] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems*, 2017.

[45] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *International Conference on Learning Representations*, 2019.

[46] J. Klicpera, A. Bojchevski, and S. Günnemann, "Predict then propagate: Graph neural networks meet personalized PageRank," in *International Conference on Learning Representations*, 2019.

[47] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations*, 2018.

[48] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?" in *International Conference on Learning Representations*, 2022.

[49] B. Donon, B. Donnot, I. Guyon, and A. Marot, "Graph neural solver for power systems," in *International Joint Conference on Neural Networks*, 2019, pp. 1–8.

[50] V. Bolz, J. Rueß, and A. Zell, "Power flow approximation based on graph convolutional networks," in *18th IEEE International Conference on Machine Learning and Applications*, 2019, pp. 1679–1686.

[51] C. Kim, T. Conrad, R. Karim, J. Oelhaf, D. Riebesel, T. Arias-Vergara, A. K. Maier, J. Jäger, and S. Bayer, "Physics-informed GNN for medium-high voltage AC power flow with edge-aware attention and line search correction operator," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2026.

[52] A. Varbella, B. Gjorgiev, and G. Sansavini, "Graph neural networks for fast contingency analysis of power systems," *IEEE Transactions on Power Systems*, 2024.

[53] G. C. Ejebe and B. F. Wollenberg, "Automatic contingency selection," *IEEE Transactions on Power Apparatus and Systems*, no. 1, pp. 97–109, 1979.

[54] E. Winston and J. Z. Kolter, "Monotone operator equilibrium networks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 10 718–10 728.

[55] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating explanations for graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.